When Does a Speech Model Know to Hand Off? Reading Competence from Hidden States for Real-Time Escalation

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Abstract

A local speech model can keep a conversation going, and a stronger expert can take over the hard questions. Calling the expert on every turn adds cost and waiting time, so the local model needs an early estimate of whether its own answer will fail. Hidden-state probes give such estimates for text models. It is not known whether this signal survives speech and duplex fine-tuning, audio input, and the timing of a native listen/speak protocol. We study these questions on frozen MiniCPM-o 4.5. We use lightweight probes to read failure signals from hidden states and separately calibrate a gate for expert handoff at answer onset. We find that failure signals can transfer from text to speech, and that readout design matters in native audio generation. On the full 240-query internal benchmark, selective expert handoff raises unweighted answer-content accuracy from 40.8% to 62.9%. The full real-time timing evaluation on the same query IDs gives mean server waiting time to first answer audio of 6.10 s with aggressive handoff, versus 9.43 s with always-escalate. P50 is 0.70 s locally and 2.87 s with aggressive handoff.

1 Introduction

As a spoken question arrives, a conversational agent must decide both when to respond and whether it can answer reliably. A local speech model can manage the exchange, while a stronger expert can help with questions beyond its competence. Routine expert calls also make locally answerable questions incur the cost and wait of a handoff. The useful intervention is selective: retain local answers when they are likely to succeed and request help when they are likely to fail. This requires a signal available at the moment the speech model begins to respond.

Native duplex models already decide when to listen and when to speak [Défossez et al., 2024, Cui et al., 2026]. That ability creates a natural place to intervene, but the decision to take the floor need not reveal whether the model knows the answer. A model can be ready to speak while lacking the knowledge or reasoning needed for a correct response. Waiting until it has finished an answer would allow a later check, but would also postpone expert help.

Text models offer a starting point. Hidden-state probes and verbal self-evaluation can predict answer correctness [Kadavath et al., 2022, Azaria and Mitchell, 2023, Kossen et al., 2024], and intermediate layers can expose information absent from the output [Orgad et al., 2025]. Text routers allocate queries across models of different cost [Chen et al., 2023, Ong et al., 2024, Varshney et al., 2026]. In native speech, the model receives an evolving audio context, and the state used for routing must be produced by its listen/speak protocol. A useful text-input signal may change after speech and duplex fine-tuning, or become inaccessible at the point where a serving system needs it.

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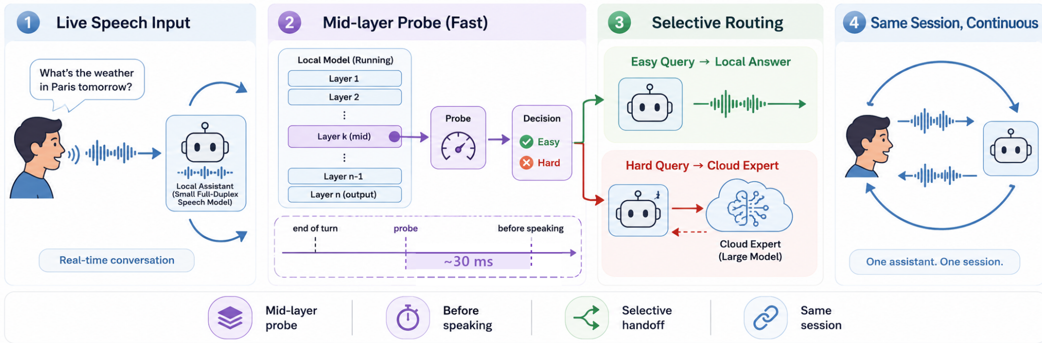

Figure 1: **The deployed system.** **(1)** Speech streams in 1 s chunks into a full-duplex local model that decides for itself when to speak. **(2)** At that speak commit, a mid-network (L22) linear read scores the turn in ∼30 ms at answer onset; the triggering call may already return the first text unit. **(3)** Lower-risk turns stay local; higher-risk turns are sent to a cloud expert while the talker stalls. **(4)** The returned answer is spoken verbatim in the talker's own voice. Unweighted answer-content accuracy on the full 240-query internal test set rises from 40.8% to 62.9% at 51.7% escalation (§4.1).

We ask whether failure information remains readable across these changes and can support an actionable handoff. Generalization matters: a probe may distinguish familiar query pools without recognizing failure on a new kind of question. Modality matters because a probe fit on text must be tested on speech-input states, including unseen human recordings. Readout design matters because the layer, position, and aggregation determine what information the gate receives. These questions connect representation analysis to deployment: a predictive score is useful only if it remains informative in the intended input regime and arrives in time to guide the response.

We study frozen MiniCPM-o [Cui et al., 2026], with raw-backbone and other speech-model controls. Our layer and position audit traces how failure readout depends on the input and calibration regime (Figure 2); a matched native ablation separates aggregation from layer choice (Figure 4). Text-trained probes also transfer to speech-input states on unseen human recordings (Figure 5). To turn this evidence into a serving decision, we separately fit an aggregated answer-onset gate on native audio features. It reuses the local model's hidden states without updating the model or waiting for a completed answer. Lower-risk turns continue locally; higher-risk turns request an expert answer that the local talker then voices.

The resulting handoff improves answer-content accuracy and, at the aggressive tier, exceeds the matched random-mixture reference. This connects readable failure information to selective expert use, while leaving an essential conversational question open: whether the improved answer is worth the interruption and wait. We report the judged answer-content benchmark alongside the full real-time timing evaluation to compare accuracy and server waiting time across routing policies. User-perceived delay and interruption quality remain to be evaluated.

2 Related Work

Competence readout. Correctness can be estimated from verbal self-evaluation [Kadavath et al., 2022, Lin et al., 2022], hidden representations [Azaria and Mitchell, 2023, Burns et al., 2023, Marks and Tegmark, 2024], and uncertainty across sampled answers [Kuhn et al., 2023, Farquhar et al., 2024]. Intermediate-layer representations can expose information missing from model outputs [Orgad et al., 2025, Mahaut et al., 2024]; Semantic Entropy Probes amortize uncertainty estimation into a hidden-state read [Kossen et al., 2024]. Our question is how readout location, input modality, and serving protocol affect this signal after speech and duplex fine-tuning.

Selective routing. FrugalGPT, HybridLLM, and RouteLLM allocate work across language models [Chen et al., 2023, Ding et al., 2024, Ong et al., 2024]; AutoMix verifies a drafted answer before escalating [Aggarwal et al., 2024]. Pre-generation activation routers and frozen-state success probes are especially close to our method [Varshney et al., 2026, Lugoloobi et al., 2026]. A linear failure probe is therefore not itself novel. We study its robustness in speech representations and calibrate thresholds for a tunable expert-call budget; local failure remains distinct from expert benefit.

Native duplex control. Native spoken-dialogue models coordinate listening and speaking [Défossez et al., 2024, Cui et al., 2026], using state or synchronization mechanisms [Wang et al., 2024a, Zhang et al., 2024, Veluri et al., 2024] or frozen-backbone adapters [Wang et al., 2024b]. BayLing-Duplex [Fang et al., 2026] makes dialogue-state decisions within a single autoregressive LLM by interleaving user audio, assistant text, and assistant audio and introducing four state tokens. Starting from GLM-4-Voice, it learns turn-taking and interruption through supervised full-duplex fine-tuning and timing-focused preference optimization. Its evaluation measures spoken-answer quality and interaction timing, providing a concrete example of learned native floor control.

Competence-aware handoff. DuplexOmni, chronological thinking, and MoshiRAG connect speech to reasoning or retrieval [Huang et al., 2026, Wu et al., 2026, Chien et al., 2026]. Our work studies a decision complementary to native floor control: given a duplex-trained checkpoint, can its hidden states predict local-answer failure and guide escalation to an expert under a tunable call budget? We fit a small readout without updating the speech model. This requires regime-specific calibration and does not assume that a learned turn-taking signal predicts answer correctness.

3 Probe-Guided Expert Handoff

3.1 The local–expert loop

Selective model routing balances response quality against the cost of a stronger model [Ong et al., 2024]. This suits our goal: improve answer accuracy under a limited expert-call budget. We adapt the routing decision to a local full-duplex speech model's transition from listening to speaking (Figure 1). The local model processes incoming speech and decides when to respond; the gate then chooses whether to continue locally or request expert help. On the expert path, an audio uplink transcribes the question, the expert returns a text answer, and the local talker voices the prepared relay text. The request and relay add cost and waiting time. We use local-answer failure risk as a proxy for escalation value; gains depend on expert correction and faithful relay of the answer.

3.2 A probe for the handoff decision

Linear *probes* test what information a classifier can read from frozen representations [Alain and Bengio, 2016]. Hidden-state classifiers have estimated statement truthfulness [Azaria and Mitchell, 2023] and pre-generation task success [Lugoloobi et al., 2026]. We use this approach to ask whether a speech model's existing states contain a directly readable failure signal. Reusing these states avoids generating a separate self-evaluation and leaves the speech model unchanged. The linear restriction tests directly accessible information without ruling out information requiring a nonlinear readout.

Offline, we collect local answers to calibration questions and use a judge to assign correctness labels. Each label is paired with hidden-state features captured at the chosen read point. We fit a logistic probe to predict failure from these features while keeping the speech model frozen. For features $\phi(x)$ from input context x, the fitted weights w, b give the risk score

$$s(x) = \sigma\big(w^\top \phi(x) + b\big)\,. \tag{1}$$

At inference, the probe reads the current features without a reference answer or an online judge. Scores above a language-specific threshold τ_ℓ favor escalation; thresholds select nominal expert-call budgets from calibration scores and stay fixed during evaluation (§4.1). Realized call rates can change when the query distribution shifts. An auxiliary dialogue-act head restricts escalation to information requests, filtering turns such as acknowledgments and stop commands (Appendix D).

3.3 Where should the probe read?

Adapting the probe to duplex speech requires choosing its layer, position, and read time. The final layer feeds the output head, but its failure signal can degrade under a change in query distribution or input modality. Figure 2 motivates L22 through text-input leave-one-pool-out (LOPO) evaluation:

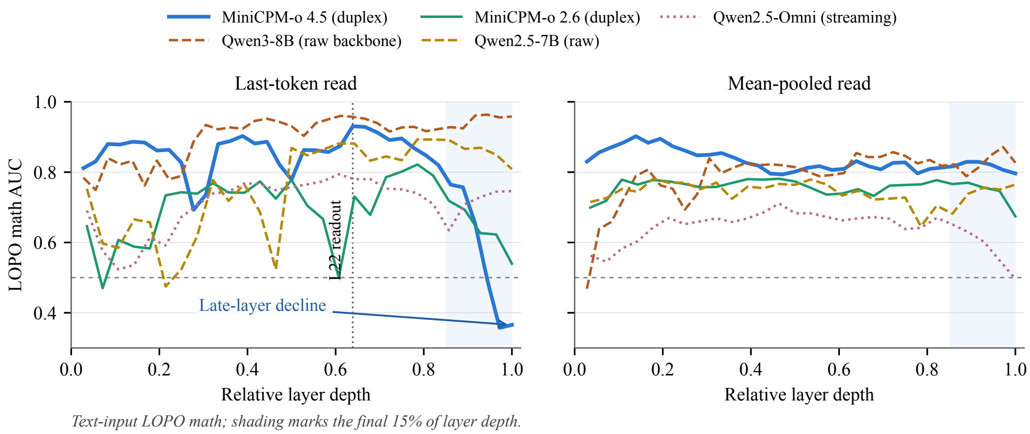

Figure 2: **Text-input transfer depends on layer and position.** The arrow and shading highlight MiniCPM-o 4.5's late-layer decline in text-input LOPO math AUC. **This decline may reflect specialization for duplex interaction control.** The model predicts a binary Listen/Speak control token before generating content [Cui et al., 2026, §3.3]. L22 is the native readout; see §4.4 and Appendix B.2 for audio-input results and mechanism controls.

on held-out math, its last-token probe reaches 0.931 AUC, versus 0.366 at the final layer. This failure does not persist when both calibration and inference use audio input (§4.4).

The native feature vector concatenates the latest L22 tail state, the mean of the tail states, and the mean input state, allowing the probe to combine the current boundary state with pooled context. The matched native ablation in §4.4 shows that this aggregation matters: a single L22 state does not outperform a single final-layer state. Both depth and aggregation therefore define the native readout.

We read these features when the generation call first returns a listen-to-speak transition. This makes the gate available at the local model's response boundary without waiting for a completed answer. The call can already contain the first generated text unit, so we call the decision point *answer onset*. Further answer chunks may expose additional failure information, at the cost of a later handoff (Appendix D.2). We fit the native probe and thresholds on features and answer labels from this serving regime to match its audio input and generation behavior. This calibration is separate from the representation-transfer study (§4.5), which alone does not provide a zero-shot native gate.

4 Experiments

4.1 Experimental setup

We evaluate frozen MiniCPM-o 4.5 (9B), with `gpt-5.5` as the escalation expert, on the full 240-query internal test set and four external English speech-QA pools: Speech TriviaQA, Speech Web Questions, and Speech Llama Questions from OpenAudioBench [Li et al., 2025], and SD-QA from VoiceBench [Faisal et al., 2021, Chen et al., 2024]. For the answer-content benchmark, the native probe is fitted on 5,221 calibration rows after excluding benchmark overlaps. Its results were rerun with this refitted probe; per-language thresholds use the remaining core rows' out-of-fold scores and stay fixed across pools. Calibration details and model controls are in Appendix A.

Each query has one sampled native session per arm: local-only, conservative, balanced, aggressive, and always-escalate. We report *answer-content accuracy*, scored from generated local text or intended relay text. The random reference is the analytical mixture $A_{\mathrm{mix}}(r) = (1-r)A_{\mathrm{local}} + rA_{\mathrm{always}}$ at the gate's realized call rate r. Both model families use all 240 Internal query IDs: chat (60), factual QA (40), knowledge (60), math (60), and SimpleQA traps (20). Every query has unit weight in accuracy and call-rate calculations. Table 9 gives the per-tier results and uncertainty intervals on this full test set. External averages weight the four QA pools equally.

Metric sources and timing endpoint. All tables and trade-off figures use the retained judged benchmark for accuracy and escalation rate, and the complete `ttfa-v3` run for latency on the same

Table 1: **Answer-content accuracy (%).** Both models use all 240 Internal IDs with unit weights and four external English QA pools; avg is the equal-pool external mean and Δ its gain over local. MiniCPM uses one native session per query/arm and fixed thresholds; random mixes endpoints at the realized call rate (Appendix D.1). NVDA is cached native-frame replay; Internal is post-selection (Appendix C).

	Internal	TriviaQA	WebQ	Llama Q.	SD-QA	avg	Δ
MiniCPM native sessions (answer-content accuracy)							
always-local	40.8	62.8	52.8	82.4	48.0	61.5	—
+ gate, conservative	46.7	60.4	59.2	78.8	54.0	63.1	+1.6
+ gate, balanced	54.2	70.0	62.4	82.4	65.0	70.0	+8.5
+ gate, aggressive	**62.9**	**88.4**	**75.2**	**85.6**	**82.5**	**82.9**	**+21.4**
matched random, aggressive	56.1	81.5	68.9	84.1	73.7	77.1	+15.6
always-escalate	70.4	96.0	79.6	91.6	87.0	88.6	+27.1
NVDA three-layer answer-onset probe (pass-3 native-frame replay)							
always-local	19.6	35.8	37.6	70.8	31.0	43.8	—
+ gate, conservative	31.7	48.4	48.8	80.7	42.0	55.0	+11.2
+ gate, balanced	44.6	61.0	59.2	85.8	54.0	65.0	+21.2
+ gate, aggressive	**60.4**	**76.8**	**69.6**	**89.7**	**66.5**	**75.7**	**+31.9**
matched random, aggressive	53.5	65.7	60.2	82.0	60.0	67.0	+23.2
always-escalate	91.7	95.5	82.8	93.1	89.0	90.1	+46.3

query IDs. This timing run contains 5,950 recorded sessions across five pools and five policies; TTFA comes directly from session timestamps. We compare the resulting per-policy summaries. Appendix A.2 gives the input and recording details for each metric. Server TTFA measures the wait from scheduled input end until the first answer PCM block is ready; audio ready earlier incurs zero wait. Candidate and stall audio are excluded; client playback is unmeasured. Appendix A.1 provides the model, probe-fitting, and answer-judging configurations used in these experiments.

4.2 Internal accuracy and timing cost

The upper MiniCPM block of Table 1 reports unweighted answer-content accuracy on the full 240-query internal test set: 40.8% locally, then 46.7%, 54.2%, and 62.9% for conservative, balanced, and aggressive. Their realized escalation rates are 8.3%, 25.0%, and 51.7% (Appendix D.1). Aggressive exceeds the matched random reference of 56.1% by 6.8 percentage points, computed before rounding. Per-tier results and accuracy intervals are in Table 9. Aggressive recovers 74.6% of the measured local-to-always accuracy gap at a call rate of 51.7%. This measures selection value relative to the recorded endpoints.

Gains vary across the fixed internal strata: factual QA improves by 52.5 points, while math accuracy is unchanged. Of 60 knowledge questions, 43 exceed the expert's input window. These descriptive results show task differences and an input constraint; they do not isolate a truncation effect on accuracy. All queries remain in the five-category analysis (Appendix A.3).

The complete `ttfa-v3` run measures server waiting time to first answer audio after input end on the same 240 query IDs (Figure 3). Internal P50 is 0.70 s locally, 1.61 s at balanced, and 2.87 s at aggressive. Aggressive has mean 6.10 s and P95 18.14 s; always has mean 9.43 s and P50 6.67 s. Local mean is 1.06 s, including 17/240 attempts (7.1%) that respond before input end and incur zero wait. Early responses do not establish better interaction or correctness. Table 10 counts failures and reports timing among completed sessions. Under the reporting convention in §4.1, aggressive has 62.9% accuracy and 6.10 s mean TTFA, compared with 70.4% and 9.43 s for always-escalate: 7.5 percentage points of accuracy are exchanged for a 35.3% reduction in mean waiting time.

4.3 External transfer and its limits

The native gate's weights, feature definition, layer, and per-language thresholds stay fixed across the four external English QA pools. The upper block's *avg* excludes the internal set; Δ is its gain over always-local. Aggressive macro accuracy rises from **61.5% to 82.9%**, compared with 77.1% for

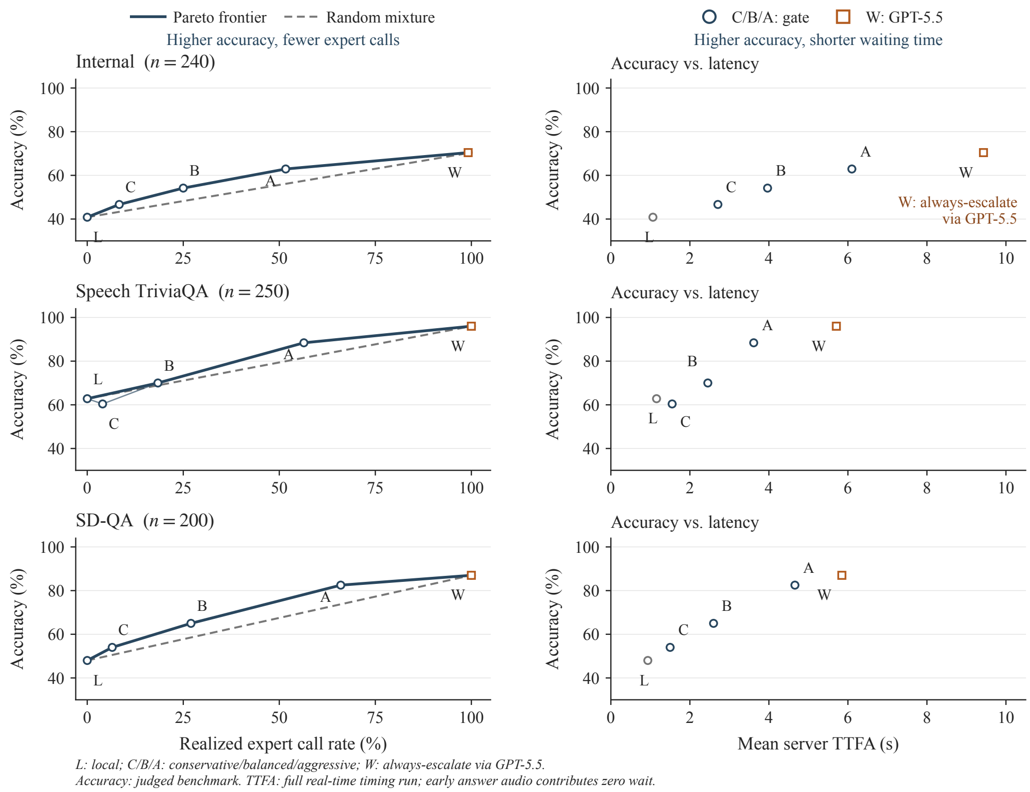

Figure 3: **Accuracy–cost trade-offs across routing policies.** Left: accuracy versus expert call rate, with thin policy curves, thick Pareto frontiers, and dashed random-mixture lines. Right: accuracy versus mean server TTFA, shown as unconnected points. Always-escalate via GPT-5.5 (W) has the highest accuracy and latency on these pools; gated policies trade some accuracy for shorter waits. TriviaQA conservative is dominated by local. Accuracy retains the judged benchmark values; TTFA uses the full timing run (§4.1). Internal retains all 240 IDs with unit weights. L is local; C/B/A are conservative/balanced/aggressive.

the reported random mixture: +21.4 points over local answering and +5.9 over random. Its realized call rate ranges from 18.8% on Llama Questions to 66.0% on SD-QA, so this is not a uniform 50%-budget comparison. TriviaQA and SD-QA illustrate large improvements over local answering (+25.6 and +34.5 points) and over the corresponding random reference (+6.9 and +8.8 points).

The lower tiers are less reliable. Conservative accuracy falls by 2.4 points on TriviaQA and 3.6 on Llama Questions; balanced leaves Llama Questions unchanged. Separately sampled sessions and limited expert headroom constrain the interpretation of small differences. The results support aggressive-tier gains across these pools, not improvement on every benchmark at every budget. On AlpacaEval, we measure 3.68 locally and 4.01 at aggressive, with a random-mixture reference of approximately 3.9; the always arm scores 4.20 versus 4.95 for returned expert text. The length cap on relayed answers is particularly restrictive for tasks requiring long, open-ended responses.

Failure categories and later reads help explain the boundary. In the native failure-type audit, specific-but-wrong answers comprise 70% of failures and are ranked at AUC 0.81, while execution slips in multi-step reasoning are near chance (0.47). In a separate read-time sweep, external mean AUC increases from 0.755 at commitment to 0.791 after two answer chunks. Later chunks expose useful failure information without identifying the mechanism underlying this change (Appendix D.2).

The lower block of Table 1 reports the exploratory NemotronLabs-VoiceChat-11B (NVDA) study. Its *avg* covers four English QA pools and excludes the Internal column. At a 50% replay budget, average accuracy rises from 43.8% to 75.7% ($\Delta = 31.9$ points), compared with 67.0% for random. This is native-frame replay with cached outcome mixtures, not another set of MiniCPM native ses-

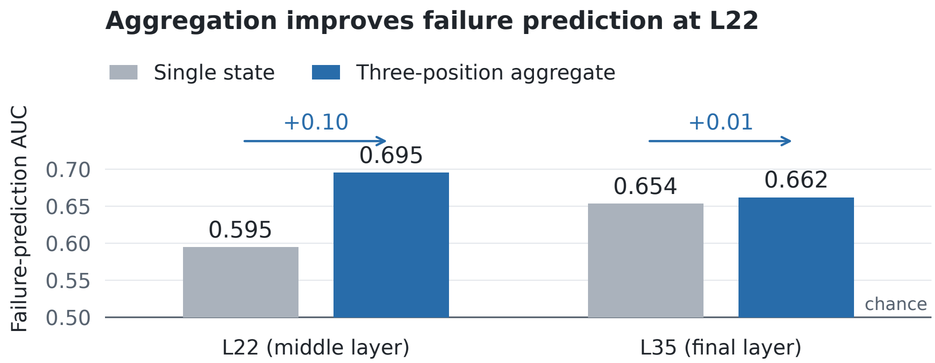

Figure 4: **Aggregation improves failure prediction at L22.** Mean AUC across five external pools under leave-one-pool-out evaluation on one native audio batch. Gray bars read the latest tail state; blue bars concatenate that state with the tail mean and input mean (§3.3). Arrows indicate the aggregation gain at each layer; the horizontal line marks chance AUC of 0.5.

sions. The L26/L30/L34 heads reach 0.818 out-of-fold AUC and 0.816 mean external AUC in the separate probe analysis (Appendix C); Table 1 reports accuracy rather than AUC. External pools were inspected during variant review, so independent confirmation is needed.

4.4 Readout ablations

Text-input transfer. We evaluate the readout without using expert outcomes. The sweep in Figure 2 compares read positions and raw-backbone controls; mean pooling reduces late-layer degradation. A raw Qwen backbone, subsampled to match per-pool failure rates, remains near 0.96, with similar ordering in the Qwen2.5 control family (Table 5). With audio calibration and audio input, the final-layer probe reaches 0.936 on LOPO math. These controls link degradation to the tested speech checkpoints and input regimes without identifying its cause (Appendix B.2).

Native layer and aggregation. We test whether predictive gains at native answer onset come from layer choice or feature aggregation. A 2×2 comparison crosses L22 versus the final layer with a single tail state versus the three-part aggregate defined in §3.3. All four readouts use one audio batch with identical training pools, labels, splits, and head recipe (Figure 4). At L22, aggregation raises mean failure-prediction AUC from 0.595 to 0.695; at the final layer, the change is smaller (0.654 to 0.662). The single-state L22 read has no advantage over the final layer. These results identify an aggregation benefit at L22, without establishing an advantage from middle-layer selection alone. Named per-pool results and cached routing gains are reported in Appendix B.4.

4.5 Text-trained probes transfer to unseen human speech

We next test whether a probe fitted to text-input hidden states can predict failure under speech input. This tests a change of input modality; Figure 2 tests held-out task-pool transfer within text input.

Figure 5 (left) holds question content fixed: the same 600 queries are presented as text and as TTS audio. The text-trained MiniCPM-o 4.5 probe reaches **0.855** AUC on audio states at L22. Transfer also appears in MiniCPM-o 2.6 and Qwen2.5-Omni, with plateau AUCs of 0.74–0.80 and 0.80–0.83, respectively; the frozen-backbone Freeze-Omni control shows much weaker transfer under the tested recipe. This panel tests modality transfer with shared question content.

The right panel tests generalization to *new questions and human speech*: probes fitted on the calibration pool are evaluated on 200 query-disjoint SD-QA recordings. The text-trained MiniCPM-o 4.5 probe reaches **0.760** AUC at L22, with a per-layer peak of 0.800 at L25. Audio-trained probes and evaluation on SD-QA text provide references for the modality change. Thus, transfer extends beyond matched TTS questions to held-out human speech. The two panels use different evaluation sets, so their AUC difference does not isolate a human-speech penalty (Appendix B.1). These results concern representation transfer; the native gate requires a separately fitted readout (§3.3).

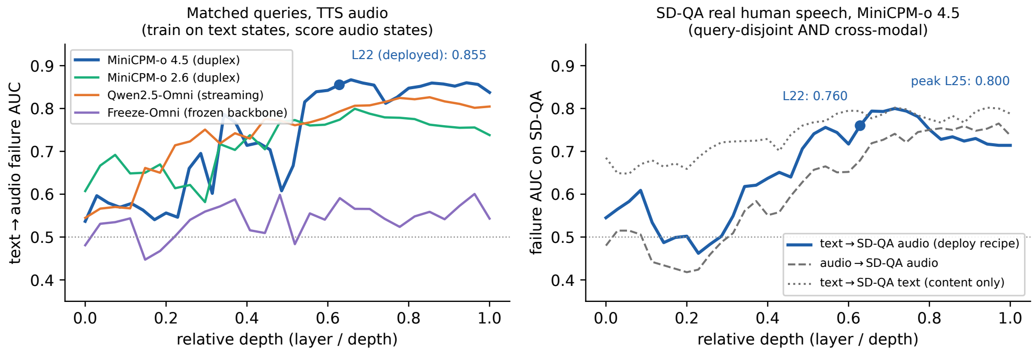

Figure 5: **Text-trained failure probes transfer to unseen human speech.** Left: text-trained probes are evaluated on TTS audio for the same 600 questions across four models; MiniCPM-o 4.5 reaches 0.855 AUC at L22. Right: MiniCPM-o 4.5 probes fitted on the calibration pool are evaluated on 200 query-disjoint SD-QA human recordings. The text-trained probe (blue) reaches 0.760 at L22 and peaks at 0.800 at L25; gray curves show audio-trained and text-target references. Both panels show relative depth, normalized by the total number of layers in each model.

5 Discussion and Limitations

Three boundaries define the scope of the results. First, coverage is limited to the tested checkpoints and predominantly single-turn speech QA. Broader conversational tasks, including multi-turn dialogue and expert continuity, remain to be evaluated. Second, native serving requires in-regime audio features and answer labels to fit the readout and set thresholds; representation transfer alone does not provide a zero-shot gate. Third, the reported latency ends at server PCM readiness. Early responses need not improve turn-taking. Human-rated speech quality and timestamps of audio playback on the client remain unmeasured in these experiments. The internal results retain all 240 queries with equal weight, but the fixed mixture is not a population estimate of conversational traffic. Task-stratified results and per-tier uncertainty appear in Tables 2 and 9.

6 Conclusion

We find that failure information remains readable in the tested frozen speech models and can transfer from text to unseen human speech. Its accessibility depends on the input regime and readout design. With native calibration, an aggregated answer-onset readout turns that information into selective expert calls and improves answer-content accuracy without updating the speech model.

The broader implication is that control of when to speak can be complemented by an estimate of when to seek help. Local failure risk provides a useful basis for that decision, although it does not by itself establish that an expert will help or that the handoff will feel worthwhile to the user. The next step is to evaluate the whole exchange: whether the answer improves, how long the user waits to hear it, and whether the system preserves turn-taking and continuity across the handoff. Such joint evaluation is needed to turn selective expert use into demonstrated conversational benefit.

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A Benchmark and calibration details

The Internal test set contains the same 240 query IDs for both model families: chat 60, factual QA 40, knowledge 60, math 60, and SimpleQA traps 20. Each query has unit weight in accuracy and call-rate calculations. All original queries remain in the reported accuracy denominator. Table 9 reports per-tier MiniCPM accuracy, uncertainty, and matched random references on this full test set. The reported timing statistics use the same 240 IDs; external pools retain their original membership. Archived Mandarin diagnostics remain separate from the four-pool English macro average.

A.1 Experimental setup details

Models and decoding. The target is MiniCPM-o 4.5 (9B, based on Qwen3-8B). Matched controls include raw backbones, a second MiniCPM generation, Qwen2.5-Omni, Qwen2-Audio, and Freeze-Omni (Table 3). Representation analyses use bf16, greedy decoding, and seed 42. Native decoding uses temperature $T = 0.7$ and top-$k = 20$. The expert is `gpt-5.5` with low reasoning effort.

Probe fitting and thresholds. Representation probes use 360 calibration queries and single-position 4,096-dimensional features. The native gate uses 5,221 calibration rows and three L22 summaries: the last tail state, the tail mean, and the input mean. Its logistic head uses $C = 3 \times 10^{-4}$. Five-fold stratified cross-validation uses shuffle seed 42. For each language, conservative, balanced, and aggressive thresholds are the $(1 - r)$ quantiles of the 4,979 core rows' out-of-fold scores, with $r \in \{0.15, 0.30, 0.50\}$. The 242 fresh training rows enter the fit but not the quantile base.

Judge configuration. Speech TriviaQA, Speech Web Questions, and Speech Llama Questions use the OpenAudioBench judge with `gpt-4o-2024-08-06`. Internal, SD-QA, and Mandarin reasoning answers use the reference-anchored `gpt-5.4-mini` adequacy judge. AlpacaEval uses VoiceBench's `gpt-4o-mini` judge on its 1–5 scale. The adequacy judge receives the question, an optional reference, and a candidate answer without the system identity. Its rubric requires factual correctness, valid reasoning, and a complete answer. Prompt details are in Appendix E.

A.2 Metric provenance and recording details

Accuracy and escalation rates retain the judged native benchmark records. In that benchmark, the expert receives up to the final 30 seconds of the prerecorded question. Server TTFA uses the complete `ttfa-v3` timing run (run ID `ttfa1`) on the same query cohorts and named policies, with one attempt per query and policy. Input is replayed in real time, expert requests use the audio prefix consumed at the trigger, and both local and relay paths produce audio. The reported accuracy values are retained from the judged benchmark; the timing run supplies the latency values. These sources are combined at the policy-summary level in the Pareto figures. The timing statistics include 5,904 completed sessions out of 5,950 attempts; 46 failed attempts have no TTFA value. The 64 completed early responses contribute zero waiting time. Candidate and stall audio do not define the answer endpoint, and client playback is unmeasured.

A.3 Internal task strata

Table 2: Descriptive strata of the full 240-query Internal test set, with unit query weights. Local, aggressive (A), and always report answer-content accuracy (%); call rate is the aggressive arm's realized rate. The final column counts question waveforms longer than the expert's 30-second input window. All five categories and all original queries are retained.

Stratum	n	Local	A	Always	A call rate	$> 30\,\mathrm{s}$ (n)
Chat	60	65.0	70.0	70.0	35.0	0
Factual QA	40	32.5	85.0	92.5	62.5	0
Knowledge	60	15.0	31.7	38.3	71.7	43
Math	60	61.7	61.7	78.3	25.0	3
SimpleQA traps	20	0.0	95.0	100.0	100.0	0

The largest gains occur on factual QA and SimpleQA traps. Math is unchanged at the aggregate level, consistent with the failure-type analysis in Appendix D.2. Long-input truncation is concentrated in the knowledge stratum (43 of 60 questions); all queries remain in the reported denominator.

B Models and representation analyses

Table 3: Model roster. "Pair" identifies a fine-tuned model and its raw backbone control.

model	type	pair	role
MiniCPM-o 4.5	omni + duplex FT	Qwen3-8B	system target
Qwen3-8B	raw	—	pair control
MiniCPM-o 2.6	duplex FT	Qwen2.5-7B	replication
Qwen2.5-Omni-7B	streaming omni FT	Qwen2.5-7B	omni control
Qwen2.5-7B	raw	—	pair control
Qwen2-Audio-7B	audio FT	Qwen2-7B	audio control
Freeze-Omni	frozen LLM + adapter	Qwen2-7B	adapter control
Qwen2-7B	raw	—	pair control

B.1 Signal comparison and layer transfer

Table 4: Failure-prediction AUC on MiniCPM-o 4.5. In-dist uses five-fold out-of-fold predictions; LOPO fits on other pools and evaluates on the named held-out query pool.

signal	in-dist	LOPO math	audio input
max entropy@4	0.696	near chance	—
$p(\mathrm{True})$-pre	0.807	trap 0.945	degrades
$p(\mathrm{True})$-post	**0.899**	—	degrades
final-layer probe	0.822	**0.372**	0.936
mid-layer (L22) probe	0.866	**0.931**	**0.855**

The final-layer state predicts source pool at 95.8% accuracy, and a pool-identity oracle alone reaches failure AUC 0.715. This motivates LOPO rather than an in-distribution comparison. The depth sweep shows that the failure is localized: mid-layer last-token reads transfer while late reads degrade on both duplex models; mean pooling degrades less. In the signal table, the final-layer audio entry is audio-calibrated LOPO math, whereas L22 tests a text-calibrated probe on speech states.

Table 5: LOPO math AUC by depth for matched model families.

model	best mid-layer	final layer	shape
Qwen3-8B	0.964 (L33/36)	0.958	plateau
MiniCPM-o 4.5	**0.931 (L22/36)**	**0.366**	late inversion
Qwen2.5-7B	0.893 (L21/28)	0.809	mild dip
Qwen2.5-Omni	0.794 (L16/28)	0.746	diffuse decline
MiniCPM-o 2.6	**0.822 (L21/28)**	**0.540**	late cliff

The text-to-audio result is not explained by synthetic speech alone: on 200 human SD-QA recordings, the text-calibrated read remains at 0.76–0.80 AUC across L22–L26. Audio-input verbal self-evaluation is less reliable; transcripts restore it, while probing needs no extra generation.

B.2 Mechanism checks

Matched controls narrow the late-layer effect without assigning it to a specific training objective. A speak-mode template leaves final-layer LOPO math unchanged (0.362 versus 0.366), the audio–text direction is nearly orthogonal to the failing probe, and source-pool identity is decodable at all depths. A residual-component analysis shows the distributed read holding through the middle layers and decaying only near the outputs of the two duplex models; it remains intact in their raw backbones and the omni-streaming control. These controls support a mid-layer read in the tested conditions, while leaving the causal role of individual fine-tuning objectives unresolved.

Direct tests further constrain the Listen/Speak hypothesis. The final probe's maximum cosine with control-token unembeddings is 0.069, versus 0.068 for the random control. A listen/speak prompt intervention on 60 calibration queries gives cue–probe cosine 0.019, near the 0.016 chance reference. These tests do not support a simple turn-control takeover of the failing probe direction; they do not rule out more distributed effects of duplex fine-tuning.

B.3 Query-surface baseline

A TF–IDF logistic router reaches 0.669 out-of-fold AUC and 0.38–0.57 under LOPO, including 0.230 on the all-failure trap pool. The hidden-state probe's advantage is therefore not explained by query wording or source identity alone. On audio labels it improves to 0.814, below L22's 0.879.

B.4 Native feature ablation

Figure 4 uses one native audio batch (agg0, official configuration), with all layer features captured in the same sessions and judged against their own generated answers. Single-state and aggregated features have 4,096 and 12,288 dimensions, respectively. Each readout fits a logistic head ($C = 3 \times 10^{-4}$) on four pools and evaluates the fifth. The five pools include Mandarin reasoning; this set differs from the main table's four English external pools. Table 6 identifies the individual pool results that underlie the averages, including the weaker reasoning scores.

The routing comparison selects the highest-risk 30% of questions in each pool and combines recorded local outcomes with cached always-arm expert outcomes. Figure 6 reports pooled accuracy gains over 2,000 matched random allocations. Per-pool gains in Table 6 use the analytical 30% random mixture. These are cached outcome comparisons, not new live gate sessions. For the L22 single-state read, 0.0505 of the random allocations matched or exceeded the observed accuracy; none did for the other three readouts. These tests compare each readout to random routing only.

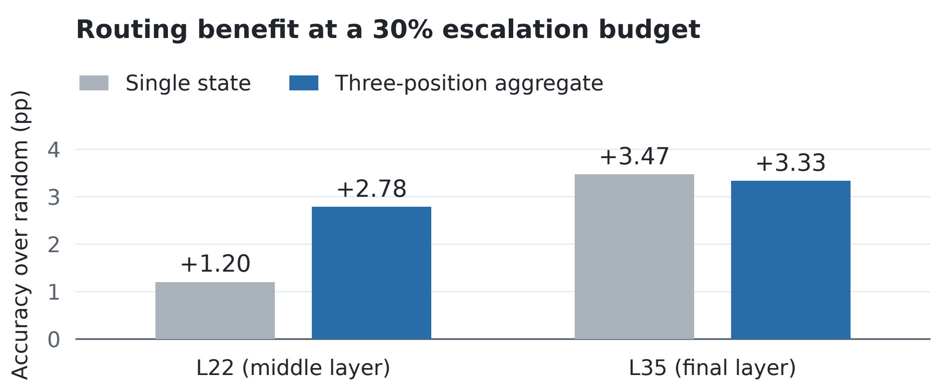

Figure 6: **Cached routing gains at a matched 30% budget.** Accuracy gains over random escalation, pooled across the same five evaluation sets as Figure 4. Each bar uses the recorded local and expert outcomes; feature choices and colors match the main ablation figure.

For context, the separately trained deployed gate reaches mean AUC 0.750 and a pooled gain of 4.80 percentage points on these rows. It was fitted on the full calibration merge, so it is not a matched training condition for the four LOPO ablations and is excluded from their plots.

C Second duplex family

We replay NemotronLabs-VoiceChat-11B through its native frame protocol with the checkpoint frozen. Of 2,550 English calibration queries, 2,481 produce a commit frame. Standardized logistic heads read L26/L30/L34 onset states; their normalized logits are averaged. The recipe is chosen on calibration-only LOPO metrics, and external labels are excluded from fitting. Each head uses

Table 6: Native ablation by evaluation pool. Each column is one of the four matched LOPO readouts in Figure 4. The lower block gives accuracy gains over random escalation in percentage points. AUC averages weight pools equally; pooled routing gains weight individual questions.

	L22		L35	
Pool	Single	Aggregate	Single	Aggregate
Failure-prediction AUC				
Speech TriviaQA	0.645	0.744	0.704	0.661
Speech Web Questions	0.579	0.732	0.646	0.683
Speech Llama Questions	0.614	0.711	0.672	0.674
SD-QA	0.648	0.713	0.682	0.669
Reasoning QA (zh)	0.488	0.576	0.564	0.624
Equal-pool mean	0.595	0.695	0.654	0.662
Accuracy gain over random at a 30% budget (pp)				
Speech TriviaQA	+3.38	+4.58	+3.78	+3.38
Speech Web Questions	+1.69	+1.69	+4.16	+3.33
Speech Llama Questions	+1.51	+3.96	+2.73	+3.96
SD-QA	+2.85	+5.35	+5.85	+4.35
Reasoning QA (zh)	-3.91	-1.93	+1.04	+1.53
Pooled gain	+1.20	+2.78	+3.47	+3.33

$C = 10^{-4}$ and concatenates the commit state, eighth post-commit state, first-eight-state mean, and running pre-commit mean (17,920 dimensions per layer). External pools were inspected during variant review, so these results provide exploratory transfer evidence. The read covers the first 640 ms after commit, so this experiment tests method transfer rather than end-to-end real-time serving.

Table 7: VoiceChat-11B transfer. Accuracy equally weights the four external English speech-QA pools; the random reference uses the same escalation rate as the probe-guided policy.

expert budget	15%	30%	50%
probe-guided accuracy	55.0	65.0	75.7
random-mixture accuracy	50.7	57.7	67.0

Internal test set. The Internal replay uses a separate fit on 2,258 committed calibration rows, excluding all 240 original internal test IDs. Its local labels judge the exact pass-3 answers; expert outcomes come from the frozen expert-answer cache. The same 240 IDs as MiniCPM contain 223 valid onset reads and 17 no-onset queries. The latter remain local under selective policies and stay in the accuracy denominator. Local and always-expert accuracy are 19.6% and 91.7%. The archived 15/30/50% scoreable-cohort policies select 33/67/112 queries, realizing 13.8/27.9/46.7% calls over all 240 and attaining 31.7/44.6/60.4% accuracy. Random escalation is restricted to this same scoreable cohort; its expected accuracy is 29.6/39.9/53.5%. These are the original full-cohort replay results. The selected layers and feature recipe were reviewed using the original Internal split, making this a post-selection diagnostic.

The three-layer read reaches 0.818 out-of-fold AUC and 0.816 mean external AUC under the pool-aligned judges. A strictly causal commit-only variant loses about 0.03 external AUC, making answer onset the closest matching read point across the two tested model families.

D Native gate and serving details

The L22 hidden state is available after roughly 57–61% of prefill. On one H100, probe availability has P50/P95 of 20/25 ms for text and 45/104 ms for audio. These numbers measure the local read, not the network or expert call. The native benchmark's realized rates are below; their variation is expected because thresholds are calibration quantiles rather than per-pool quotas.

Dialogue-act filtering. A second linear head uses the same L22 read to distinguish information requests from floor-management turns. Escalation requires both the information-seeking score $a(x)$ and failure score $s(x)$ to exceed their respective thresholds in the same turn:

$$d(x) = \mathbf{1}\{s(x) \geq \tau_\ell\}\,\mathbf{1}\{a(x) \geq \tau_a\}. \tag{2}$$

On the scripted calibration stimuli, the act head has OOF AUC 1.000; the question-side 0.5th-percentile threshold removes 0.5% of true escalations and admits no floor-management stimuli.

D.1 Realized escalation rates

Table 8: Realized escalation rates (%) for MiniCPM native results in Table 1, plus the archived Mandarin and separate AlpacaEval pools. Internal retains all 240 queries; all pools use unit query weights. These differ from nominal calibration and NVDA replay budgets. Local-only never escalates.

Pool	n	Conservative	Balanced	Aggressive	Always
Internal	240	8.3	25.0	51.7	99.2
Speech TriviaQA	250	4.0	18.4	56.4	100.0
Speech WebQ	250	4.8	22.0	60.0	99.2
Speech Llama Q.	250	0.4	3.6	18.8	100.0
SD-QA	200	6.5	27.0	66.0	100.0
Reasoning QA (zh)	202	19.8	33.2	50.5	100.0
AlpacaEval	199	1.5	4.5	43.7	100.0

Table 9: Internal results on all 240 original test queries (%), with unit query weights. Random mixes the measured local/always endpoints at the corresponding call rate. Accuracy has pointwise 95% percentile intervals from 100,000 paired query-bootstrap resamples (seed 42), conditional on the fixed test set and recorded arms. The intervals exclude judging and model-rerun uncertainty.

tier	calls	accuracy [95% CI]	random
local-only	0.0	40.8 [34.6,47.1]	—
conservative	8.3	46.7 [40.4,52.9]	43.3
balanced	25.0	54.2 [47.9,60.4]	48.2
aggressive	51.7	62.9 [56.7,68.8]	56.1
always	99.2	70.4 [64.6,76.3]	—

Table 10: Internal server TTFA (seconds), complete `ttfa-v3` run on all 240 test IDs: 240 attempts per arm, with 239/238/236/236/237 completed and 1/2/4/4/3 failed in row order. TTFA is the non-negative wait from scheduled input end to first answer PCM. All completed sessions count; the 17/12/11/13/7 early responses have zero wait. Failed attempts have no TTFA value. Timing ends at the first local PCM-bearing chunk return or relay PCM yield, excluding candidate/stall audio and client playback.

tier	mean	P50	P95	P99
local-only	1.06	0.70	2.62	3.64
conservative	2.71	0.89	10.85	34.08
balanced	3.97	1.61	14.41	26.91
aggressive	6.10	2.87	18.14	54.78
always	9.43	6.67	25.86	60.83

D.2 Failure type and later reads

This separate sweep fits a probe at each read point using 5,236 judged calibration rows, row-random five-fold fitting, and $C = 3 \times 10^{-4}$. Its Internal column retains the original 240-query cohort, as in the main accuracy table, but uses this sweep's separately fitted readouts. Later reads pad finished answers with the last available vector. Its five-pool macro includes Mandarin reasoning, unlike the main table's four-English-pool macro. Failure categories are assigned by a model judge from recorded answers.

Failure-prediction AUC at successive native read points. External macro equally weights five speech-QA pools.

Read point	Calibration OOF	Internal	Reasoning-zh	External macro
Before onset	.848	.832	.711	.753
At onset	.850	.852	.648	.755
After 1 chunk	.852	.854	.681	.762
After 2 chunks	.855	.846	.769	.791
After 3 chunks	.857	.858	.744	.801

Specific-but-wrong answers account for about 70% of failures and are ranked at AUC 0.813; perception errors reach 0.889, while execution slips are near chance at 0.469. The later-read gains on reasoning expose a tradeoff between earlier routing and information that emerges during generation.

E Judge prompt and reproducibility

The answer judge receives the question, an optional reference, and a candidate answer without the system identity. It marks the candidate adequate only when it is factually correct, uses valid reasoning, and completely answers the question; partial, hedged, refused, off-topic, and truncated answers are inadequate. The pre-answer self-evaluation asks the model not to answer the question and to reply exactly "Yes" or "No" to whether it would answer correctly. The post-draft version presents the question and proposed answer and asks whether the answer is correct.

Representation-study sampling and splits use seed 42. The native benchmark uses sampled decoding and retains one run per query and arm. MiniCPM-o 4.5 runs with PyTorch 2.8.0 and Transformers 4.51.0 on single-H100 instances. The release includes the calibration manifest, out-of-fold scores, thresholds, overlap exclusions, benchmark hashes, and scripts that rebuild the figures and accuracy and timing summaries from the original recorded data for these experiments.